

Ronghua Xu, Ph.D.

✉ rxu22@binghamton.edu

🌐 <https://scholar.google.com/citations?user=gKf0U28AAAAJ&hl=en>

🐙 <https://github.com/samuelxu999>

🌐 <https://www.linkedin.com/in/ronghua-xu-bu/>



Biographical Sketch

About me

- ◇ Ronghua Xu is an Tenure-Track Assistant Professor of Applied Computing at Michigan Technological University. He earned a Ph.D. and an M.S. in Electrical and Computer Engineering at the Binghamton University - State University of New York (SUNY) in 2023 and 2018 respectively. He also received an M.S. in Mechanical and Electrical Engineering from Nanjing University of Aeronautics & Astronautics in 2010 and a B.S. in Mechanical Engineering from Nanjing University of Science & Technology, China in 2007. Before joining Binghamton University, he worked at Siemens on software development, system integration, and test automation from June 2010 to June 2016.

Research Interests

- ◇ Blockchain and Distributed Ledger Technology, Internet-of-Things (IoT), Machine Learning (ML), Cloud/Fog/Edge Computing Paradigm.
- ◇ Blockchain and smart contract enabled security solutions to Internet of Things (IoTs)
- ◇ Intelligence, assurance and resilience of next generation network.

Education

- | | |
|---------------------|---|
| Jun 2018 – Aug 2023 | ◇ Ph.D., Electrical and Computer Engineering , Binghamton University-SUNY, Binghamton, NY, USA.
Dissertation title: <i>A Secure-by-Design Federated Microchain Fabric for Internet-of-Things(IoT) System</i>
Advisor: Prof. Yu Chen |
| Aug 2016 – May 2018 | ◇ MS, Computer Engineering , Binghamton University-SUNY, Binghamton, NY, USA.
Thesis title: <i>Capability Based Access Control Strategies to Deter DDoS Attacks Exploiting IoT Devices</i>
Advisor: Prof. Yu Chen |
| Sep 2007 – Mar 2010 | ◇ MS, Mechanical and Electrical Engineering , Nanjing University of Aeronautics & Astronautics, Nanjing, China.
Thesis title: <i>Research on Form-to-function Mapping and Re-creative Design Method Based on Function Ontology</i>
Advisor: Prof. Dunbing Tang |
| Sep 2003 – Jul 2007 | ◇ BS, Mechanical Engineering , Nanjing University of Science & Technology, Nanjing, China. |

Employment History

- | | |
|---------------------------|---|
| August 2023 – Now | ◇ Tenure-Track Assistant Professor. Michigan Technological University, MI, USA. |
| August 2018 – August 2023 | ◇ Graduate Researcher. Binghamton University-SUNY, NY, USA. |
| May 2021 – August 2021 | ◇ Technical Intern. PSE System & Hardware Division, ZF North America, Inc., MI, USA. |

Employment History (continued)

June 2010 – June 2016 ♦ **Software Engineer.** Department of Software Development, Research & Development Division, Siemens Numerical Control Ltd., Nanjing, China.

Teaching Experience

Instructor	♦ Internet of Medical Things (IoMT) and Remote Patient Monitoring (RPM) (SAT5317) , Spring 2025
	♦ Blockchain Fundamentals and Applications (SAT5980) , Spring 2024
	♦ Digital Forensics (SAT4816-5816) , Fall 2023/2024.
Teaching Assistant	♦ Digital Logic Design (EECE-251) , Fall 2018, 2019.
	♦ Sophomore Design (EECE-287) , Spring 2019, 2020.
Guest Lecturer	♦ Senior Design I (EECE 487) , Fall 2022.
	♦ Computer Network Architecture (EECE-453/553) , Fall 2018 - 2022.
	♦ Network Security (EECE-658) , Spring 2018 - 2019.

Skills

Languages	♦ Strong reading, writing and speaking competencies for English, Mandarin Chinese.
Coding	♦ C/C++, Java, Python, C#, VB, tclsh, bash, powershell, SQL, XML/XSL, $\LaTeX$, ...
Databases	♦ MySQL, PostgreSQL, SQLite.
Web Dev	♦ HTML, css, JavaScript, Flask Web Server, Figma UI Design.
Misc.	♦ Academic research, teaching, training, consultation.

Research Publications

Journal Articles

- 1 Liu, X., & **Xu, R.** (2025a). Bimw: Blockchain-enabled innocuous model watermarking for secure ownership verification. *Future Internet*, 17(11), 490. [doi:10.3390/fi17110490](https://doi.org/10.3390/fi17110490)
- 2 Liu, X., & **Xu, R.** (2025b). From vulnerability to robustness: A survey of patch attacks and defenses in computer vision. *Electronics*, 14(23), 4553. [doi:10.3390/electronics14234553](https://doi.org/10.3390/electronics14234553)
- 3 Liu, X., **Xu, R.**, & Chen, Y. (2024). A decentralized digital watermarking framework for secure and auditable video data in smart vehicular networks. *Future Internet*, 16(11), 390. [doi:10.3390/fi16110390](https://doi.org/10.3390/fi16110390)
- 4 Liu, X., **Xu, R.**, & Zhao, C. (2024). Agfi-gan: An attention-guided and feature-integrated watermarking model based on generative adversarial network framework for secure and auditable medical imaging application. *Electronics*, 14(1), 86. [doi:10.3390/electronics14010086](https://doi.org/10.3390/electronics14010086)
- 5 Qu, Q., Hatami, M., **Xu, R.**, Nagothu, D., Chen, Y., Li, X., ... Chen, G. (2024). The microverse: A task-oriented edge-scale metaverse. *Future Internet*, 16(2), 60. [doi:10.3390/fi16020060](https://doi.org/10.3390/fi16020060)
- 6 **Xu, R.**, Nagothu, D., Chen, Y., Aved, A., Ardiles-Cruz, E., & Blasch, E. (2024). A secure interconnected autonomous system architecture for multi-domain iot ecosystems. *IEEE Communications Magazine*, 62(7), 52–57. [doi:10.1109/MCOM.001.2300354](https://doi.org/10.1109/MCOM.001.2300354)
- 7 Nagothu, D., **Xu, R.**, Chen, Y., Blasch, E., & Aved, A. (2022). Deterring deepfake attacks with an electrical network frequency fingerprints approach. *Future Internet*, 14(5), 125. [doi:10.3390/fi14050125](https://doi.org/10.3390/fi14050125)
- 8 **Xu, R.**, & Chen, Y. (2022a). μ Dfl: A secure microchained decentralized federated learning fabric atop iot networks. *IEEE Transactions on Network and Service Management*. [doi:10.1109/TNSM.2022.3179892](https://doi.org/10.1109/TNSM.2022.3179892)

- 9 **Xu, R.**, Chen, Y., Chen, G., & Blasch, E. (2022). Sausa: Securing access, usage, and storage of 3d point cloud data by a blockchain-based authentication network. *Future Internet*, 14(12), 354. [doi:10.3390/fi14120354](https://doi.org/10.3390/fi14120354)
- 10 **Xu, R.**, Wei, S., Chen, Y., Chen, G., & Pham, K. (2022). Lightman: A lightweight microchained fabric for assurance-and resilience-oriented urban air mobility networks. *Drones*, 6(12), 421. [doi:10.3390/drones6120421](https://doi.org/10.3390/drones6120421)
- 11 Qu, Q., **Xu, R.**, Chen, Y., Blasch, E., & Aved, A. (2021). Enable fair proof-of-work (pow) consensus for blockchains in iot by miner twins (mint). *Future Internet*, 13(11), 291. [doi:10.3390/fi13110291](https://doi.org/10.3390/fi13110291)
- 12 **Xu, R.**, Nagothu, D., & Chen, Y. (2021a). Decentralized video input authentication as an edge service for smart cities. *IEEE Consumer Electronics Magazine*, 10(6), 76–82. [doi:10.1109/MCE.2021.3062564](https://doi.org/10.1109/MCE.2021.3062564)
- 13 **Xu, R.**, Nagothu, D., & Chen, Y. (2021b). Econledger: A proof-of-enf consensus based lightweight distributed ledger for iotv networks. *Future Internet*, 13(10), 248. [doi:10.3390/fi13100248](https://doi.org/10.3390/fi13100248)
- 14 **Xu, R.**, Nikouei, S. Y., Nagothu, D., Fitwi, A., & Chen, Y. (2020). Blendsp: A blockchain-enabled decentralized smart public safety system. *Smart Cities*, 3(3), 928–951. [doi:10.3390/smartcities3030047](https://doi.org/10.3390/smartcities3030047)
- 15 **Xu, R.**, Chen, Y., Blasch, E., & Chen, G. (2019). Exploration of blockchain-enabled decentralized capability-based access control strategy for space situation awareness. *Optical Engineering*, 58(4), 041609. [doi:10.1117/1.OE.58.4.041609](https://doi.org/10.1117/1.OE.58.4.041609)
- 16 **Xu, R.**, Chen, Y., Blasch, E., & Chen, G. (2018c). Blendcac: A smart contract enabled decentralized capability-based access control mechanism for the iot. *Computers*, 7(3), 39. [doi:10.3390/computers7030039](https://doi.org/10.3390/computers7030039)

Conference Proceedings

- 1 Liu, X., **Xu, R.**, & Peng, X. (2025). Bewsat: Blockchain-enabled watermarking for secure authentication and tamper localization in industrial visual inspection. In *Eighth international conference on machine vision and applications (icmva 2025)* (Vol. 13734, pp. 54–65). SPIE.
- 2 Ogunbunmi, S., **Xu, R.**, & Chen, Y. (2025). A lightweight microchained architecture for unmanned aerial vehicle network reputation systems. In *Proceedings of the 2025 7th blockchain and internet of things conference* (pp. 15–25). ACM.
- 3 **Xu, R.**, Liu, X., Nagothu, D., Qu, Q., & Chen, Y. (2025). Detecting manipulated digital entities through real-world anchors. In *International conference on advanced information networking and applications* (pp. 450–461). Springer.
- 4 Nagothu, D., **Xu, R.**, & Chen, Y. (2023). Dema: Decentralized electrical network frequency map for social media authentication. In *Disruptive technologies in information sciences vii* (Vol. 12542, pp. 57–72). SPIE.
- 5 Nagothu, D., **Xu, R.**, Chen, Y., Blasch, E., & Ardiles-Cruz, E. (2023). Application of electrical network frequency as an entropy generator in distributed systems. In *Naecon 2023-ieee national aerospace and electronics conference* (pp. 233–238). IEEE.
- 6 Ogunbunmi, S., Hatmai, M., **Xu, R.**, Chen, Y., Blasch, E., Ardiles-Cruz, E., ... Chen, G. (2023). A lightweight reputation system for uav networks. In *International conference on security and privacy in cyber-physical systems and smart vehicles* (pp. 114–129). Springer.
- 7 Qu, Q., **Xu, R.**, Sun, H., Chen, Y., Sarkar, S., & Ray, I. (2023). A digital healthcare service architecture for seniors safety monitoring in metaverse. In *2023 ieee international conference on metaverse computing, networking and applications (metacom)* (pp. 86–93). IEEE.
- 8 Wei, S., Huang, H., Chen, G., Blasch, E., Chen, Y., **Xu, R.**, & Pham, K. (2023). Rodad: Resilience oriented decentralized anomaly detection for urban air mobility networks. In *2023 integrated communication, navigation and surveillance conference (icns)* (pp. 1–11). IEEE.

- 9 **Xu, R.**, & Chen, Y. (2022b). Fairledger: A fair proof-of-sequential-work based lightweight distributed ledger for iot networks. In *2022 ieee international conference on blockchain (blockchain)* (pp. 348–355). IEEE. [doi:10.1109/Blockchain55522.2022.00055](https://doi.org/10.1109/Blockchain55522.2022.00055)
- 10 **Xu, R.**, Chen, Y., Li, X., & Blasch, E. (2022). A secure dynamic edge resource federation architecture for cross-domain iot systems. In *2022 international conference on computer communications and networks (icccn)* (pp. 1–7). IEEE. [doi:10.1109/ICCCN54977.2022.9868843](https://doi.org/10.1109/ICCCN54977.2022.9868843)
- 11 Nagothu, D., **Xu, R.**, Chen, Y., Blasch, E., & Aved, A. (2021a). Defake: Decentralized enf-consensus based deepfake detection in video conferencing. In *2021 ieee 23rd international workshop on multimedia signal processing (mmsp)* (pp. 1–6). IEEE. [doi:10.1109/MMSP53017.2021.9733503](https://doi.org/10.1109/MMSP53017.2021.9733503)
- 12 Nagothu, D., **Xu, R.**, Chen, Y., Blasch, E., & Aved, A. (2021b). Detecting compromised edge smart cameras using lightweight environmental fingerprint consensus. In *Proceedings of the 19th acm conference on embedded networked sensor systems* (pp. 505–510). ACM. [doi:10.1145/3485730.3493684](https://doi.org/10.1145/3485730.3493684)
- 13 **Xu, R.**, & Chen, Y. (2021). Fed-ddm: A federated ledgers based framework for hierarchical decentralized data marketplaces. In *2021 international conference on computer communications and networks (icccn)* (pp. 1–8). IEEE. [doi:10.1109/ICCCN52240.2021.9522359](https://doi.org/10.1109/ICCCN52240.2021.9522359)
- 14 Qu, Q., **Xu, R.**, Nikouei, S. Y., & Chen, Y. (2020). An experimental study on microservices based edge computing platforms. In *Ieee infocom 2020-ieee conference on computer communications workshops (infocom wkshps)* (pp. 836–841). IEEE. [doi:10.1109/INFOCOMWKSHPS50562.2020.9163068](https://doi.org/10.1109/INFOCOMWKSHPS50562.2020.9163068)
- 15 **Xu, R.**, Chen, Y., Blasch, E., Aved, A., Chen, G., & Shen, D. (2020). Hybrid blockchain-enabled secure microservices fabric for decentralized multi-domain avionics systems. In *Sensors and systems for space applications xiii* (Vol. 11422, 114220J). International Society for Optics and Photonics. [doi:10.1117/12.2559036](https://doi.org/10.1117/12.2559036)
- 16 **Xu, R.**, Chen, Y., & Li, J. (2020). Poster: Microfl: A lightweight, secure-by-design edge network fabric for decentralized iot systems. In *The network and distributed system security symposium (ndss)*. Retrieved from https://www.ndss-symposium.org/wp-content/uploads/2020/02/NDSS2020posters_paper_19.pdf
- 17 **Xu, R.**, Zhai, Z., Chen, Y., & Lum, J. K. (2020). Bit: A blockchain integrated time banking system for community exchange economy. In *2020 ieee international smart cities conference (isc2)* (pp. 1–8). IEEE. [doi:10.1109/ISC251055.2020.9239045](https://doi.org/10.1109/ISC251055.2020.9239045)
- 18 Blasch, E., **Xu, R.**, Nikouei, S. Y., & Chen, Y. (2019). A study of lightweight dddas architecture for real-time public safety applications through hybrid simulation. In *2019 winter simulation conference (wsc)* (pp. 762–773). IEEE. [doi:10.1109/WSC40007.2019.9004727](https://doi.org/10.1109/WSC40007.2019.9004727)
- 19 Lin, X., **Xu, R.**, Chen, Y., & Lum, J. K. (2019). A blockchain-enabled decentralized time banking for a new social value system. In *2019 ieee conference on communications and network security (cns)* (pp. 1–5). IEEE. [doi:10.1109/CNS.2019.8802734](https://doi.org/10.1109/CNS.2019.8802734)
- 20 Nikouei, S. Y., **Xu, R.**, Chen, Y., Aved, A., & Blasch, E. (2019). Decentralized smart surveillance through microservices platform. In *Sensors and systems for space applications xii* (Vol. 11017, 110170K). International Society for Optics and Photonics. [doi:10.1117/12.2518999](https://doi.org/10.1117/12.2518999)
- 21 **Xu, R.**, Chen, S., Yang, L., Chen, Y., & Chen, G. (2019). Decentralized autonomous imaging data processing using blockchain. In *Multimodal biomedical imaging xiv* (Vol. 10871, pp. 72–82). SPIE. [doi:10.1117/12.2513243](https://doi.org/10.1117/12.2513243)
- 22 **Xu, R.**, Nikouei, S. Y., Chen, Y., Blasch, E., & Aved, A. (2019). Blendmas: A blockchain-enabled decentralized microservices architecture for smart public safety. In *2019 ieee international conference on blockchain (blockchain)* (pp. 564–571). IEEE. [doi:10.1109/Blockchain.2019.00082](https://doi.org/10.1109/Blockchain.2019.00082)
- 23 **Xu, R.**, Ramachandran, G. S., Chen, Y., & Krishnamachari, B. (2019). Blendsm-ddm: Blockchain-enabled secure microservices for decentralized data marketplaces. In *2019 ieee international smart cities conference (isc2)* (pp. 14–17). IEEE. [doi:10.1109/ISC246665.2019.9071766](https://doi.org/10.1109/ISC246665.2019.9071766)

- 24 Nagothu, D., **Xu, R.**, Nikouei, S. Y., & Chen, Y. (2018). A microservice-enabled architecture for smart surveillance using blockchain technology. In *2018 IEEE International Smart Cities Conference (ISC2)* (pp. 1–4). IEEE. [doi:10.1109/ISC2.2018.8656968](https://doi.org/10.1109/ISC2.2018.8656968)
- 25 Nikouei, S. Y., Chen, Y., Song, S., **Xu, R.**, Choi, B.-Y., & Faughnan, T. (2018). Smart surveillance as an edge network service: From harr-cascade, svm to a lightweight cnn. In *2018 IEEE 4th International Conference on Collaboration and Internet Computing (CIC)* (pp. 256–265). IEEE. [doi:10.1109/CIC.2018.00042](https://doi.org/10.1109/CIC.2018.00042)
- 26 Nikouei, S. Y., Chen, Y., Song, S., **Xu, R.**, Choi, B.-Y., & Faughnan, T. R. (2018). Real-time human detection as an edge service enabled by a lightweight cnn. In *2018 IEEE International Conference on Edge Computing (Edge)* (pp. 125–129). IEEE. [doi:10.1109/EDGE.2018.00025](https://doi.org/10.1109/EDGE.2018.00025)
- 27 Nikouei, S. Y., **Xu, R.**, Nagothu, D., Chen, Y., Aved, A., & Blasch, E. (2018). Real-time index authentication for event-oriented surveillance video query using blockchain. In *2018 IEEE International Smart Cities Conference (ISC2)* (pp. 1–8). IEEE. [doi:10.1109/ISC2.2018.8656668](https://doi.org/10.1109/ISC2.2018.8656668)
- 28 **Xu, R.**, Chen, Y., Blasch, E., & Chen, G. (2018a). A federated capability-based access control mechanism for internet of things (IOTs). In *Sensors and systems for space applications xi* (Vol. 10641, 106410U). International Society for Optics and Photonics. [doi:10.1117/12.2305619](https://doi.org/10.1117/12.2305619)
- 29 **Xu, R.**, Chen, Y., Blasch, E., & Chen, G. (2018b). Blendcac: A blockchain-enabled decentralized capability-based access control for IOTs. In *2018 IEEE International Conference on Internet of Things (IThings) and IEEE Green Computing and Communications (GreenCom) and IEEE Cyber, Physical and Social Computing (CPSCom) and IEEE Smart Data (SmartData)* (pp. 1027–1034). IEEE. [doi:10.1109/Cybermatics_2018.2018.00191](https://doi.org/10.1109/Cybermatics_2018.2018.00191)
- 30 **Xu, R.**, Lin, X., Dong, Q., & Chen, Y. (2018). Constructing trustworthy and safe communities on a blockchain-enabled social credits system. In *Proceedings of the 15th EAI International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services* (pp. 449–453). [doi:10.1145/3286978.3287022](https://doi.org/10.1145/3286978.3287022)
- 31 **Xu, R.**, Nikouei, S. Y., Chen, Y., Polunchenko, A., Song, S., Deng, C., & Faughnan, T. R. (2018). Real-time human objects tracking for smart surveillance at the edge. In *2018 IEEE International Conference on Communications (ICC)* (pp. 1–6). IEEE. [doi:10.1109/ICC.2018.8422970](https://doi.org/10.1109/ICC.2018.8422970)

Book Chapters

- 1 Chen, Y., & **Xu, R.** (2024b). Introductory chapter: Edge computing in the evolution of smart cities. In *Edge computing architecture-architecture and applications for smart cities* (pp. 1–11). [doi:10.5772/intechopen.1007810](https://doi.org/10.5772/intechopen.1007810)
- 2 **Xu, R.**, Nagothu, D., & Chen, Y. (2024). Ar-edge: Autonomous and resilient edge computing architecture for smart cities. (pp. 1–21). [doi:10.5772/intechopen.1005876](https://doi.org/10.5772/intechopen.1005876)
- 3 **Xu, R.**, Nagothu, D., & Chen, Y. (2023). Ecom: Epoch randomness-based consensus committee configuration for IOT blockchains. In *Principles and practice of blockchains* (pp. 135–154). [doi:10.1007/978-3-031-10507-4_7](https://doi.org/10.1007/978-3-031-10507-4_7)
- 4 **Xu, R.**, Chen, Y., & Blasch, E. (2021). Microchain: A light hierarchical consensus protocol for IOT systems. In *Blockchain applications in IOT ecosystem* (pp. 129–149). Springer.
- 5 Nagothu, D., **Xu, R.**, Nikouei, S. Y., Zhao, X., & Chen, Y. (2020). Smart surveillance for public safety enabled by edge computing. In *Edge computing: Models, technologies and applications* (pp. 409–433). [doi:10.1049/PBPC033E_ch19](https://doi.org/10.1049/PBPC033E_ch19)
- 6 **Xu, R.**, Chen, Y., & Blasch, E. (2020). Decentralized access control for IOT based on blockchain and smart contract. In *Modeling and design of secure internet of things* (pp. 505–528). [doi:10.1002/9781119593386.ch22](https://doi.org/10.1002/9781119593386.ch22)
- 7 Nikouei, S. Y., **Xu, R.**, & Chen, Y. (2019). Smart surveillance video stream processing at the edge for real-time human objects tracking. In *Fog and edge computing: Principles and paradigms* (pp. 319–346). [doi:10.1002/9781119525080.ch13](https://doi.org/10.1002/9781119525080.ch13)

Books

- 1 Chen, Y., & **Xu, R.** (2024a). *Edge computing architecture-architecture and applications for smart cities*.  doi:10.5772/intechopen.1002562
- 2 **Xu, R.**, Chen, Y., & Blasch, E. (2023). *Lightweight blockchain for internet of things: Rationale and a case study*. Bellingham, Washington 98227-0010 USA: SPIE Press.

Professional Services

Editorial Community

- ◇ Springer Nature Editorial Board - Discover Internet of Things (Jan 2025 - Now)

Conference Technical Program Committee (TPC)

- ◇ The 6th ACM International Workshop on BLockchain-enabled Networked Sensor Systems (BlockSys-24) (in conjunction with SenSys-2023).
- ◇ IEEE Fourth Intelligent Cybersecurity Conference (ICSC2024/2025).
- ◇ IEEE/ACM International Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE 2024/2025).
- ◇ The 2nd International Workshop on Decentralized Physical Infrastructure Network (DePIN 2024/2025).
- ◇ The 29th IEEE Symposium on Computers and Communications (ISCC 2024).
- ◇ The 2024 EAI International Conference on Security and Privacy in Cyber-Physical Systems and Smart Vehicles (SmartSP 2024)
- ◇ The 7th IEEE International Conference on Blockchain (Blockchain-2024).
- ◇ The 6th IFIP International Internet of Things (IoT) Conference (IFIP-IoT 2023)
- ◇ The 5th ACM International Workshop on BLockchain-enabled Networked Sensor Systems (BlockSys-23) (in conjunction with SenSys-2023).
- ◇ The 2023 EAI International Conference on Security and Privacy in Cyber-Physical Systems and Smart Vehicles (SmartSP 2023)
- ◇ The 6th IEEE International Conference on Blockchain (Blockchain-2023).
- ◇ Artificial Intelligence and Machine Learning Technologies for IoT (AMT) (IEEE WiMob-SPPDT'2023).
- ◇ The 6th International Workshop on BLockchain Enabled Sustainable Smart Cities (BLESS 2023) (in conjunction with ICCCN 2023 Conference).
- ◇ The 4th ACM International Workshop on BLockchain-enabled Networked Sensor Systems (BlockSys-22) (in conjunction with SenSys-2022).
- ◇ The 5th IEEE International Conference on Blockchain (Blockchain-2022).
- ◇ The 5th International Workshop on BLockchain Enabled Sustainable Smart Cities (BLESS 2022) (in conjunction with ICCCN 2022 Conference).
- ◇ WiMob Short Papers, Posters and Demos Track (IEEE WiMob-SPPDT'2022).
- ◇ The 4th IEEE International Conference on Blockchain (Blockchain-2021).
- ◇ The 4th International Workshop on BLockchain Enabled Sustainable Smart Cities (BLESS 2021) (in conjunction with ICCCN 2021 Conference).
- ◇ The 3rd IEEE International Conference on Blockchain (Blockchain-2020).
- ◇ The 3rd International Workshop on BLockchain Enabled Sustainable Smart Cities (BLESS 2020) (in conjunction with ISC2 2020 Conference).
- ◇ The 2nd International Workshop on BLockchain Enabled Sustainable Smart Cities (BLESS 2019) (in conjunction with ISC2 2019 Conference).

Professional Services (continued)

- ◇ The 1st International Workshop on Lightweight Blockchain for Edge Intelligence and Security (LightChain 2019).

Reviewer for Journals

- ◇ Elsevier Computer Communications
- ◇ Elsevier Computer Networks
- ◇ Elsevier Computers & Security
- ◇ Elsevier Pervasive and Mobile Computing
- ◇ Elsevier Blockchain: Research and Applications
- ◇ Elsevier Sustainable Cities and Society
- ◇ Elsevier International Journal of Intelligent Networks
- ◇ IEEE Access
- ◇ IEEE Internet-of-Things Journal (IoT-J)
- ◇ IEEE Transactions on Big Data (TBD)
- ◇ IEEE Transactions on Industrial Informatics (TII)
- ◇ IEEE Transactions on Dependable and Secure Computing (TDSC)
- ◇ IEEE Transactions on Network Science and Engineering (TNSE)
- ◇ MDPI Applied Sciences
- ◇ MDPI Sensor and Actuator Networks
- ◇ Hindawi Wireless Communications and Mobile Computing

Reviewer for Conferences

- ◇ IEEE International Conference on Computer Communications (INFOCOM)
- ◇ IEEE International Conference on Blockchain (Blockchain)
- ◇ IEEE Global Communications Conference (GLOBECOM)
- ◇ IEEE International Conference on Wireless and Mobile Computing, Networking And Communications (WiMob)
- ◇ IEEE International Performance Computing and Communications Conference (IPCCC)
- ◇ IEEE International Conference on Consumer Electronics (ICCE)
- ◇ IEEE International Conference on Communications (ICC)
- ◇ IEEE International Smart Cities Conference (ISC2)
- ◇ IEEE International Conference on Cloud Networking (CloudNet)
- ◇ ACM Conference on Embedded Networked Sensor Systems (SenSys)
- ◇ EAI SECURECOMM

Miscellaneous Experience

Awards and Achievements

- 2025 ◇ **Conference: CI PAOS: Challenges and Opportunities of Research Data Management in Construction Engineering**, (Award No. 2541825, Co-PI), NSF.
- 2024 ◇ **ICC Rapid Seeding Awards**, Michigan Technological University.
- 2023 ◇ **Graduate Student Excellence Award in Research**, Graduate School, Binghamton University.
- 2019 ◇ **2019 Computers Best Paper Award**, Multidisciplinary Digital Publishing Institute (MDPI).

Miscellaneous Experience (continued)

2018 ◇ **Outstanding MS Research**, Department of Electrical and Computer Engineering, Binghamton University.

On campus Services

◇ **Fall 2018 Leadership Volunteers**, International Student and Scholar Services (ISSS), Binghamton University.

Membership

◇ IEEE

◇ ACM